

Chapter 1

1.2 (a) Nominal, (b) Ordinal, (c) Ordinal (d) Nominal, (e) Nominal, (f) Ordinal.

1.3 (a) Let X be a student's number of correct answers on the exam, then $X \sim \text{Bin}\left(100, \frac{1}{4}\right)$.

(b) $\mu = np = 25, \sigma^2 = np(1-p) = 18.75, \sigma = \sqrt{18.75} = 4.33$.

$P(X \geq 50) = P\left(Z \geq \frac{50-25}{4.33}\right) = P(Z \geq 5.77) \approx 0$. Since the probability is nearly zero, so it's surprising if the student made at least 50 correct responses.

1.5 Refer to Exercise 1.4,

Y = number who support an increase. π = population proportion who support an increase.

(a) Suppose $y = 0$ for $n = 2$. MLE of $\pi = \hat{\pi} = \frac{y}{n} = \frac{0}{2} = 0$. This estimate seems unbelievable since the sample size is too small, and zero probability means the event will not happen, which is impossible.

(b) The Bayesian estimator is $\frac{y+1}{n+2}$. In this case, $\pi^* = \frac{0+1}{2+2} = \frac{1}{4} = 0.25$.

1.7 (a) $\left(\frac{5}{6}\right)^6$. (b) The probability that the gun doesn't fire in the first $y-1$ rounds is $\left(\frac{5}{6}\right)^{y-1}$, and the probability that the gun fires with the next trigger is just $\frac{1}{6}$. So, the probability of the gun firing in round y is just $\left(\frac{5}{6}\right)^{y-1} \times \frac{1}{6}$, for $y = 1, 2, 3, \dots$

1.9 (a) Let π = population proportion with greater relief. $\begin{cases} H_0: \pi = 0.5 \\ H_1: \pi \neq 0.5 \end{cases}$. $\hat{\pi} = 0.6$.

$Z = \frac{0.6-0.5}{\sqrt{\frac{0.5 \times 0.5}{100}}} = 2$, p -value = 0.0455 < $\alpha = 0.05$, there's enough evidence to claim that there's significant

difference in relief between the standard and new analgesic.

(b) 95% wald confidence interval for π is $\hat{\pi} \pm 1.96 \times \sqrt{\frac{\hat{\pi}(1-\hat{\pi})}{n}} = 0.6 \pm 1.96 \times \sqrt{\frac{0.6(1-0.6)}{100}} = (0.504, 0.696)$.

95% score confidence interval for π is solving $\frac{0.6-\pi_0}{\sqrt{\frac{\pi_0(1-\pi_0)}{n}}} = \pm 1.96 \Rightarrow (0.502, 0.691)$.

Note: In section 1.6 (P.19), the author demonstrated how to find confidence interval by R.

```
...
binom.confint(60,100, methods = c("asymptotic","wilson"))
x  n mean  lower  upper
60 100  0.6 0.5039818 0.6960182
60 100  0.6 0.5020026 0.6905987
...
```

$$1.12 \begin{cases} H_0: \pi = 0.5 \\ H_1: \pi \neq 0.5 \end{cases}, \hat{\pi} = \frac{0}{25} = 0.$$

(a) When we conduct Wald test, $Z = \frac{\hat{\pi} - \pi_0}{\sqrt{\frac{\hat{\pi}(1-\hat{\pi})}{n}}} = \frac{0-0.5}{\sqrt{\frac{0(1-0)}{25}}} = \frac{-0.5}{0}$ (no meaning).

(b) The 95% wald confidence interval for π is $\hat{\pi} \pm 1.96 \times \sqrt{\frac{\hat{\pi}(1-\hat{\pi})}{n}} = 0 \pm 1.96 \times \sqrt{\frac{0(1-0)}{25}} = 0 \pm 0 = (0, 0)$.

This confidence interval is not believable.

(c) When we conduct score test, $Z = \frac{\hat{\pi} - \pi_0}{\sqrt{\frac{\pi_0(1-\pi_0)}{n}}} = \frac{0-0.5}{\sqrt{\frac{0.5(1-0.5)}{25}}} = \frac{-0.5}{0.1} = -5 < z_{0.05/2} = -1.96$, so we reject H_0 .

(d) Since the sample size is too small, it's not recommended to use normal approximation.

$$z = \frac{\hat{\pi} - \pi_0}{\sqrt{\pi_0(1-\pi_0)}} \Rightarrow \pm 1.96 = \frac{0 - \pi_0}{\sqrt{\pi_0(1-\pi_0)}} (+ \text{ has no solution}) \Rightarrow \pi_0 \approx 0.133.$$

Thus, the score confidence interval for π is (0,0.133).

Note: In section 1.6 (P.19), the author demonstrated how to find confidence interval by R.

...

```
binom.confint(0,25, conf.level = 0.95, methods = "wilson")
```

...

Output:

```
method x  n mean lower      upper
wilson 0 25    0    0 0.1331923
```

1.15 (a) $Var(Y) = n\pi(1-\pi)$. Let $\hat{\pi} = \frac{Y}{n}$. $Var(\hat{\pi}) = Var\left(\frac{Y}{n}\right) = \frac{1}{n^2} Var(Y) = \frac{1}{n^2} \times n\pi(1-\pi) = \frac{\pi(1-\pi)}{n}$.

$$\therefore \sigma(\hat{\pi}) = \sqrt{\frac{\pi(1-\pi)}{n}}.$$

(b) Since $\sigma(\hat{\pi})$ is maximized when $\pi = 0.5$, and $\sigma(\hat{\pi})$ is minimized at $\pi = 0$ or 1 , so it's more easily to estimate π precisely when it is near 0 or 1 than when it is near 0.50.

1.19 The posterior interval can be derived using the following code:

...

```
>> qbeta(c(0.025,0.975), shape1 = 0.5, shape2 = 25.5)
[1] 1.944577e-05 9.468276e-02
```

...

The interpretation can be referred to Section 1.5.3 (P.16):

(a) the frequentist interpretation of the score confidence interval (0.0, 0.133) is: If we used this method over and over with separate, independent samples, in the long run 95% of the confidence intervals would contain π .

(b) the Bayesian interpretation of the posterior interval (0.00002, 0.0947) is: After observing the data, we can say that the probability is 0.95 that π falls between 0.00002 and 0.0947.